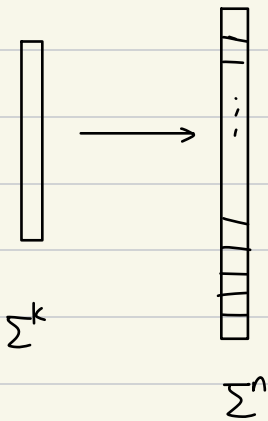
# Large Alphabet to Small Alphabet




Reed-Solomon : $\Sigma = \mathbb{F}_q$. $(q \geq n)$

Large distance : $n - k + 1$ (Singleton bound)

Turn into binary ? (In computers, e.g)

First idea : write $\Sigma$ as length $\log_2 |\Sigma|$ bits.

blocklength : $n \log_2 |\Sigma|$

Distance ? Same distance $d$.

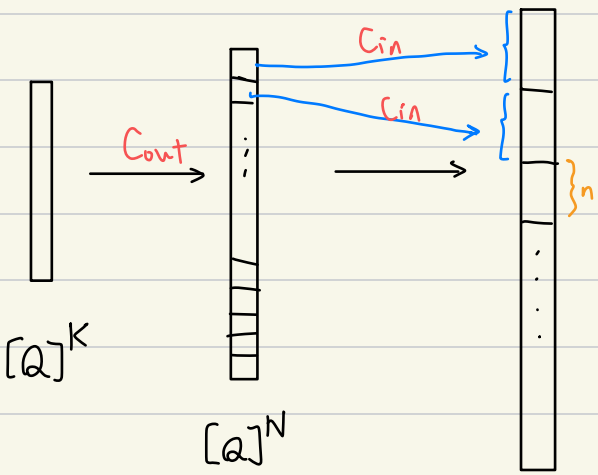
Relative dist : $\frac{\delta}{\log_2 |\Sigma|}$.

↓ 9 vs 13

1001 1101

Better idea: encode each $\sigma \in \Sigma$ using a binary code.

$\Rightarrow$ Any $\sigma \neq \sigma' \in \Sigma$ differ in many bits.




"Concatenation"

$C_{out} : [Q]^K \to [Q]^N$

$C_{in} : [q]^k \to [q]^n$

$\star\ Q = q^k$ bijection between $[Q]$ and $[q]^k$.

$\underline{C_{out} \circ C_{in}}$

Given message $m \in [Q]^K = [q]^{kK}$

$$(C_{out} \circ C_{in})(m) = \left( C_{in}(C_{out}(m)_1), \ldots, C_{in}(C_{out}(m)_N) \right).$$

In terms of $q$:

- Message length: $kK$
- Blocklength: $nN$
- Distance ?

- Linear if both are linear.

**Thm** $Q = q^k$

$C_{out}$: $(N, K, D)_Q$ code rate $R$, relative dist $\delta_{out}$

$C_{in}$: $(n, k, d)_q$ code rate $r$ $\delta_{in}$

then $C_{out} \circ C_{in}$ is an $(nN, kK, dD)_q$ code.

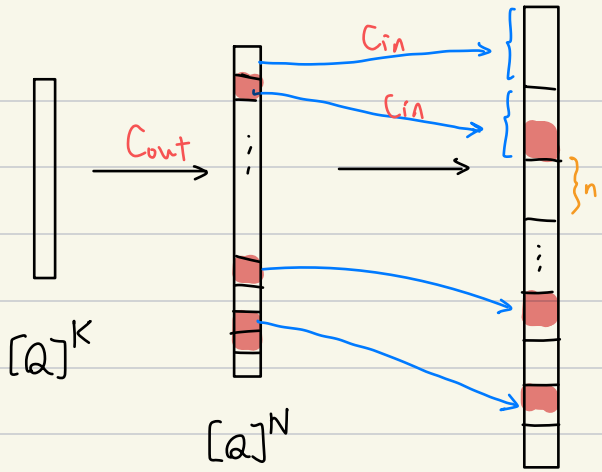
rate $rR$, rel. distance $\delta_{in} \cdot \delta_{out}$

**Pf** $m_1 \neq m_2 \in [Q]^K$.

$C_{out}(m_1)$, $C_{out}(m_2)$ differ in $\geq D$ entries.

These entry differ in $d$ locations

In total, $dD$ entries differ.




## Zyablov Bound

What's the rate-distance trade-off for concatenated codes?

Cout: large alphabet. Use Reed-Solomon!

$$\delta_{out} = 1 - R \quad \text{(Singleton)}$$

Cin: $q$ can be small.

$(r \geq 1 - H_q(\delta))$

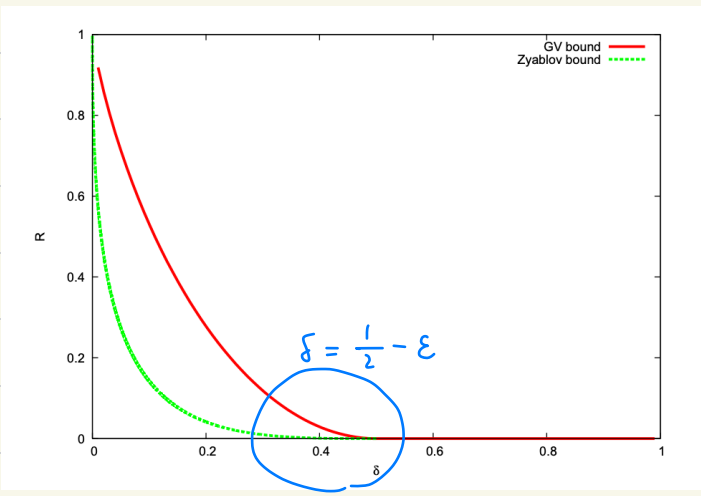
$$\text{GV}: \quad \delta_{in} = H_q^{-1}(1-r)$$

Then, $$\delta = \delta_{in} \cdot \delta_{out} = (1-R) \cdot H_q^{-1}(1-r)$$

Rate in terms of $\delta$?

$$R = 1 - \frac{\delta}{H_q^{-1}(1-r)}$$

$$\text{Rate} = \max_{r>0} \; r\left(1 - \frac{\delta}{H_q^{-1}(1-r)}\right)$$




More instructive to look at binary ($q=2$),

$$\delta = \tfrac{1}{2} - \varepsilon$$

GV: $r = \Omega(\varepsilon^2)$

Zyablov: distance $(1 - r_{out}) \cdot \left(\frac{1}{2} - \varepsilon_{in}\right) \to \frac{1}{2} - \varepsilon$

rate $r_{out} \cdot \Omega(\varepsilon_{in}^2) \to \Omega(\varepsilon^3)$.

How can we assume $C_{in}$ achieves GV bound?

$C_{out}$: Reed-Solomon

$Q = N+1$ very large!

$C_{in}$: $[q]^k \longrightarrow [q]^n$ $q$ is a constant!

$Q = q^k \Rightarrow k = \log_q Q$.

$C_{in}$ is a very "small" code!

Only $\Theta(\log N)$ length.

Explicit construction: allow poly$(N)$ time.

Brute-force search over $C_{in}$:

time $q^{kn} \approx q^{O(k^2)} = N^{O(\log N)}$.

Exercise 4.6: $q^{O(n)} = \text{poly}(N)$ time.

# Justesen Code

Why use the same inner codes ?

1. Pick $N$ different inner codes $C_{in}^1 \cdots C_{in}^N$.
2. "Most" (not necessarily all) meet the GV bound.

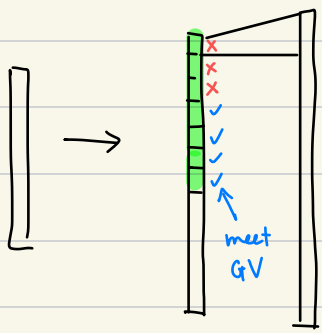
$m \in [Q]^K$, let $c = C_{out}(m)$,

$$C_{out} \circ (C_{in}^1, \cdots, C_{in}^N)(m) = (C_{in}^1(c_1), \ldots, C_{in}^N(c_N)).$$

Ensemble of codes:

At least $(1-\varepsilon)N$ of the inner codes meet the GV bound.

Pf




Distance $\geq (\delta_{out} - \varepsilon) N \cdot d_{GV}$

# Wozencraft Ensemble

For $\alpha \in \mathbb{F}_{q^k}^*$, $\left(Q = q^k \text{ many}\right)$

$C_{in}^{\alpha} : \mathbb{F}_q^k \longrightarrow \mathbb{F}_q^{2k}$ is defined as

$$C_{in}^{\alpha}(x) = (x, \alpha x).$$

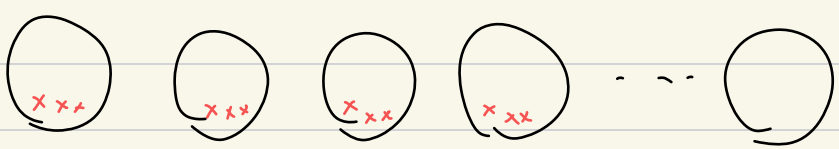
($\alpha x$: extension-field multiplication)

Rate $\frac{1}{2}$.

Goal: Show that most of them meet the GV bound.

Claim $\{C_{in}^{\alpha}\}$ all disjoint!

$\forall\, y \in \mathbb{F}_q^{2k} \setminus \{0\}$, $y \in C_{in}^{\alpha}$ for at most 1 $\alpha \in \mathbb{F}_{q^k}^*$.



Pf ① $y_1 \neq 0$, $y_2 \neq 0$: $\alpha = \frac{y_1}{y_2}$.

② $y_1 \neq 0$, $y_2 = 0$: Can't happen.

Code Constructions we've seen:

- Random
- Polynomials (RS)
- Concatenation

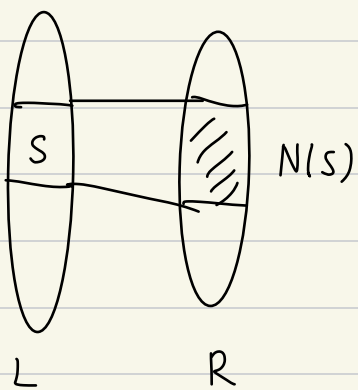
Combinatorial

# Expanders

$G = (V, E)$

Bipartite graphs $(L, R, E)$.

left right




Vertex expander: $\forall S \subseteq L, \quad |S| \leq \gamma |L|$

$$|N(S)| \geq \alpha |S|.$$

Easy: connect everything?

Want small degree! c, d regular